

M. Sc. Physics (Semester-II) 2026

Quantum Mechanics II – Home-Work

- Write the matrix form of J_x, J_y, J_z for $j = 1/2$ in the basis $|j, m\rangle$.
 - How these matrices are related to Pauli matrices?
- For Pauli matrices, evaluate explicitly by matrix multiplication the matrices $\sigma_x^2, \sigma_y^2, \sigma_z^2$. Also show that $\sigma_i \sigma_j = \delta_{ij} \mathbb{I} + i \epsilon_{ijk} \sigma_k$. Find the dimension of $\mathbb{I}$. Here δ_{ij} is Kronecker delta and ϵ_{ijk} is Levi-Civita symbol.